Happiest Health
Digital Twin
CHALLENGE
Build. Predict. Reimagine Healthcare.

BUILD A VIRTUAL PATIENT MODEL

Predict health outcomes with EHR & wearable data.

REIMAGINING & REFORMING HEALTHCARE IN INDIA

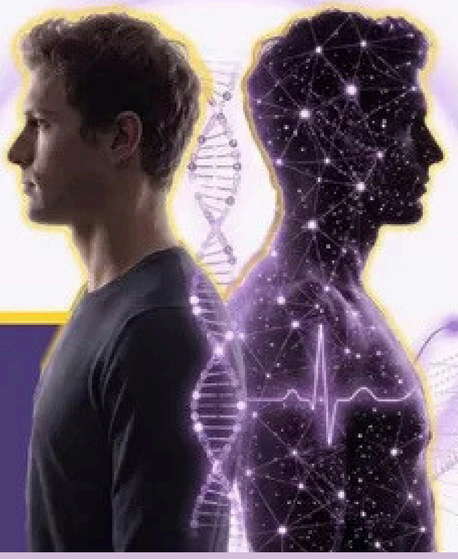
HAPPIEST HEALTH
Better Knowledge. Better Health.



VIRTUAL PATIENT DATA

AI INSIGHTS. OUTCOMES.

PREDICTIVE CARE



₹ Prize Pool of 2 LAKH

Opportunity to INTERN with us

PHASE 1 · PROTOTYPE & CODE SUBMISSION

Glucose Twin

From reactive glucose monitoring to predictive health intelligence

An AI-powered digital twin that fuses patient records with CGM, meal, activity and wearable data to forecast glucose 30, 60 and 120 minutes ahead.

TEAM

StratX

CHALLENGE

Happiest Health Digital Twin Challenge 2026

FOCUS CONDITION

Diabetes · glucose control

Why Glucose Needs a Digital Twin

Today’s glucose monitoring mostly shows what is happening now, not what comes next.

101 M

Indians living with diabetes (11.4% prevalence)

136 M

more have prediabetes and may progress to diabetes

Reactive monitoring

Highs and lows are usually noticed after they start, when the current reading already crosses a limit.

Fragmented context

Glucose history, meals, activity, wearables and patient records sit in separate places.

Little time to act

Without a forecast, patients and care teams get no early window to prevent a spike or a drop.

Our answer: a Glucose Twin that forecasts glucose 30, 60 and 120 minutes ahead and flags threshold crossings before they show up in current readings.

Source: ICMR–INDIAB study, The Lancet Diabetes & Endocrinology (2023).

Proposed solution

Glucose Twin:

From Reactive Glucose Monitoring to Predictive Health Intelligence

Glucose Twin is an AI-powered predictive health monitoring prototype that combines historical glucose readings with incoming or simulated sensor data and available meal, activity, and wearable information to forecast future glucose trends. Using an XGBoost-based prediction engine, it estimates glucose levels at 30, 60, and 120 minutes, analyses potential high- or low-threshold crossings, and presents forecasts, trends, and alerts through a unified dashboard.

CORE CONCEPT

- Predictive Visibility: View estimated future glucose trends instead of relying solely on current readings.
- Multi-Horizon Forecasting: Generate predictions for 30, 60, and 120 minutes ahead.
- Early Risk Awareness: Identify potential threshold crossings from the predicted glucose trajectory.
- Unified Health Context: Bring glucose history and available meal, activity, and wearable data into one workflow.
- Real-Time Visualisation: Present readings, forecasts, and alerts through an integrated dashboard.

Our Team Explains the Solution

Watch the video

Value Proposition

- We help users move beyond reactive glucose monitoring by forecasting future glucose trends and highlighting potential risks before they appear in current readings.

Primary Users

- People monitoring glucose levels
- Caregivers supporting glucose management
- Diabetes care teams and clinicians
- Healthcare technology providers
- Remote patient monitoring (RPM) platforms

Use Case

- Glucose Twin helps users and healthcare teams track glucose trends, predict fluctuations, and detect potential high or low thresholds through a unified dashboard integrating glucose history, meals, activity, and wearable data for proactive monitoring.

User Journey

01 **Connect Data**
Access historical glucose readings and available patient context.

02 **SYNC READINGS**
Receive incoming or simulated glucose data with timestamps.

03 **PROCESS & ANALYSE**
Validate readings and extract relevant glucose trends and features.

04 **FORECAST GLUCOSE**
The model estimates glucose levels at periodic intervals .

05 **IDENTIFY POTENTIAL RISKS**
Analyse predicted trends for possible high- or low-threshold crossings.

06 **VISUALISE INSIGHTS**
View current readings, future glucose trends, and potential alerts on the dashboard.

Two Data Streams, One Virtual Patient

Static patient records and dynamic sensor streams are fused into one view of the patient.

STATIC · HISTORICAL

Patient profile & history

- Age, sex, BMI & demographics
- Medical history & past diagnoses
- Labs: HbA1c, fasting glucose

DYNAMIC · REAL-TIME

Sensor & lifestyle streams

- CGM glucose with timestamps
- Wearables: HR, activity, sleep
- Meals: carbs, protein, fat, timing

FEATURE FUSION

- Lagged glucose readings
- Rolling statistics
- Rate of change & time of day
- Meal & activity features
- Patient context features

Leakage-safe: only data available at prediction time

OUTPUT

Virtual patient state

- Forecasts at +30, +60, +120 min
- Prediction intervals
- Threshold-crossing alerts
- Alert timeline & patient state
- Doctor-facing dashboard

Sandbox data only

CGMacros (PhysioNet): 45 participants, about 10 days each, with CGM, meals, wearable and lab data. No real patient data.

GLUCOSE TWIN: TECHNICAL ARCHITECTURE

Multimodal patient and sensor data → leakage-safe feature pipeline → XGBoost forecasts (+30, +60, +120 min) → threshold-crossing analysis → FastAPI and Next.js dashboard.

PATIENT & SENSOR INPUT DATA

1. **CGM sensor readings**
   Glucose values + timestamps (continuous / intermittent)
2. **Wearable signals**
   Heart rate, activity, sleep (optional)
3. **Meal information**
   Carbs, calories, protein, fat, meal timing
4. **Patient profile & history**
   Age, medical history, lab results, relevant context
5. **Historical dataset**
   CGMacros (and other datasets) for model training & evaluation

INGESTION & FEATURE ENGINEERING

DATA INGESTION LAYER

- CSV / Historical Files
- Simulated Live Stream
- REST API / WebSockets

PREPROCESSING (PANDAS / NUMPY)

- Data cleaning & validation
- Timestamp alignment
- Missing-value handling
- Outlier handling

FEATURE EXTRACTION PIPELINE

- Lagged glucose readings
- Rolling statistics
- Rate of change, time-of-day
- Meal and activity features
- Patient context features

ML PREDICTION ENGINE (XGBOOST)

MULTI-HORIZON GLUCOSE FORECAST



Forecast horizons

+ 30 min | + 60 min | + 120 min

XGBoost regression

scikit-learn ecosystem integration

Separate models per horizon

or multi-output model, saved via Joblib

Report model confidence

and uncertainty (e.g. prediction intervals)

FORECAST & ALERT ANALYSIS

Compare predicted trajectory

with configured high / low thresholds.

Estimate threshold-crossing

time only when supported by forecast.

ALERT CLASSIFICATION

HIGH / LOW RISK
Threshold crossing likely (within forecast horizon)

WATCH
Sparse or uncertain forecast

NORMAL
No crossing predicted in forecast horizon

BACKEND API & PERSISTENCE

FastAPI inference service

Runs preprocessing, model inference and alert analysis

REST endpoints

Predictions, historical data, patient profile & settings

WebSocket(real-time)

Push live readings, forecasts and alerts to dashboard

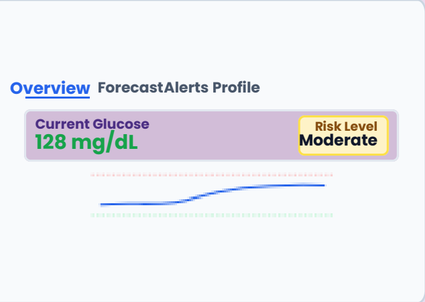
PostgreSQL (planned)

Store profiles, readings, forecasts, alerts & metadata

Auth & Access Control

User authentication & role-based access control

NEXT.JS DASHBOARD & DIGITAL TWIN



- Current glucose reading
- +30 / +60 / +120 min forecasts
- Alert timeline & crossing time
- Patient state visualization
- Optional 3D Digital Twin

Technology Stack & Model Evaluation

An open-source stack and an evaluation plan designed to avoid data leakage.

Frontend
Next.js / React

Styling
Tailwind CSS

Charts & 3D
Recharts / Spline

ML model
XGBoost
scikit-learn

Data processing
Pandas, NumPy

Real-time
WebSockets

Backend
Python FastAPI

Database
PostgreSQL

Deployment
Docker (optional)

How we will measure it

- Held-out participant and time-based split
- MAE and RMSE at +30, +60 and +120 min
- Threshold-event precision, recall and false-alert rate
- Comparison with a persistence baseline
- Strict temporal validation to prevent leakage
- Prediction intervals reported for uncertainty

Doctor Dashboard Concept

How a clinician interacts with the virtual patient (illustrative view with simulated data).



1 Review

Check current glucose, risk level and the patient profile.

2 Anticipate

Inspect +30, +60 and +120 min forecasts with their uncertainty.

3 Decide

Use the alert timeline and crossing time to plan follow-up.

Impact & Benefits

Earlier awareness for patients, one connected view for care teams.

101 M

Indians living with diabetes

120 min

forecast look-ahead window

5

input streams fused into one view

3

alert states: High/Low risk, Watch, Normal

People with diabetes

See likely highs and lows ahead of time and plan around meals and activity.

Caregivers

Clear alerts and forecasts make day-to-day glucose support easier.

Clinicians & care teams

One view of glucose history, forecasts, risk level and patient context.

Health-tech & RPM platforms

API-first design (REST + WebSocket) that plugs into remote monitoring.

Source for 101 M: ICMR–INDIAB study, The Lancet Diabetes & Endocrinology (2023).

PoC Outcomes & Roadmap

What this proof of concept delivers today, and how it can grow.

What the proof of concept delivers

- XGBoost glucose forecasts at +30, +60 and +120 minutes
- Fusion of static patient context with dynamic CGM, meal and wearable data
- Threshold-crossing alert engine with uncertainty-aware logic
- Conceptual doctor dashboard for the virtual patient
- Evaluation against a persistence baseline: MAE, RMSE, precision, recall

Roadmap

1

Proof of concept — NOW

Forecasting model, alert logic and dashboard on public and synthetic data.

2

Harden the platform — NEXT

PostgreSQL storage, authentication, role-based access and calibrated prediction intervals.

3

Validate in the real world — LATER

Test on real CGM and wearable data under ethical approval; pilot with care teams.

Thank You

From reactive monitoring to predictive health intelligence.

Forecast

Glucose at +30, +60 and +120 minutes with XGBoost

Fuse

Patient records with CGM, meal, activity and wearable data

Alert

Likely highs and lows before they show in current readings

Watch our demo video

Scan the QR code or open the link

Watch the video